

Wenlan Zhang

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Geospatial data scientist and final-stage PhD researcher at UCL working at the intersection of urban climate, human mobility and GeoAI. Develops reproducible spatial and machine-learning workflows integrating satellite, street-view, mobility and administrative data to study how environmental and policy shocks shape urban exposure, movement and settlement patterns. Research spans Earth Observation, multimodal visual AI, spatial econometrics and causal inference across cities in Africa, Asia and Latin America, with outputs used by government and multilateral partners. Current research interests include urban heat and climate-sensitive mobility in rapidly changing cities.

Core Competencies

- **Urban Climate & Environmental Analytics:** Earth Observation and remote-sensing-derived environmental indicators; Google Earth Engine; natural-hazard and climate-risk analysis; flood-exposure assessment; LiDAR-based environmental modelling.
- **Urban Mobility & Human–Environment Interactions:** Mobile-phone mobility data, origin-destination inference, migration and displacement, spatial interaction analysis, causal inference.
- **GeoAI & Multisource Urban Sensing:** Satellite and street-view imagery, multimodal vision-language models, computer vision, large-scale image analysis and multisource data integration.
- **Spatial Modelling & Urban Analytics:** Geographically Weighted Regression (GWR), spatial regression (SEM/SLM), spatial autocorrelation, spatial non-stationarity, comparative cross-city analysis.
- **Programming & Reproducibility:** Python (GDAL, Rasterio, GeoPandas, PyTorch, Scikit-learn, HuggingFace), R, QGIS, Google Earth Engine, SQL, Git/GitHub.

Selected Research Experience

- **Centre for Advanced Spatial Analysis (CASA), UCL** **London, UK**
PhD Researcher – Selected Independent Projects *2022–Present*
 - Developed a city-scale multimodal GeoAI pipeline (computer vision + vision-language model) across 306k street-view images to extract and map urban environmental conditions in Nairobi, Kenya, and collected supplementary field-based imagery there to test whether self-collected data improved model accuracy over Street View alone; integrated spatial clustering and 100m indicators, and open-sourced the resulting dataset through [IDEAmaps](#).
 - Leading an 18-city comparative study using digital trace mobility data to quantify how crisis events reshape urban mobility and whether impacts are disproportionately concentrated among vulnerable populations; the first stage audits representational biases in the underlying mobility data before impact estimation.
 - Built a [Google Earth Engine web application](#) for wildfire detection and environmental monitoring in Euboea, Greece, linking satellite-derived indicators with forest-management questions.
- **University of Oxford, Leverhulme Centre for Demographic Science** **Oxford, UK**
Research Assistant (Part-time) *May 2025–Present*
 - Apply causal inference to quantify measurement distortion in cross-border mobility data and evaluate how policy shocks reshape migration flows, integrating remote-sensing and administrative data sources (EU-funded Mapineq Project).
 - Investigated how neighbourhood characteristics shape residential place preferences using UK-wide survey data, linking individual preferences with geospatial indicators of the local built and social environment (BBC-funded The National Conversation project).
 - Co-authored [Gaza Tents](#), an automated computer-vision pipeline mapping 265k+ tents from satellite imagery in Gaza for displacement monitoring; the Palestinian Central Bureau of Statistics uses the restricted-tier data for resource allocation.

- **International Telecommunication Union** **Geneva, Switzerland (Remote)**
Machine Learning Research Intern *Feb 2025-May 2025*
 - Evaluated spatial generalisation across Kenya vs. Mexico; applied Geographically Weighted Regression to address spatial non-stationarity, boosting local R^2 to 0.92.
 - Benchmarked XGBoost/LightGBM against linear baselines, achieving a peak test R^2 of 0.88 using high-dimensional proxy features (OpenCellID, Nighttime Lights).
- **World Bank Group, Development Impact Evaluation (DIME)** **Washington, DC, US**
Research Assistant *Mar 2024-Dec 2024*
 - Co-developed NetMob2024, a privacy-preserving mobility dataset converting mobile-app location and trajectory data into population-density and origin-destination matrices across four LMICs (Colombia, India, Indonesia and Mexico); co-authored the accompanying dataset paper (World Bank Data Catalog) and coordinated the international data challenge and conference.
 - Conducted quasi-experimental analysis of public-transport interventions using mobile-phone trajectories and origin-destination networks to quantify changes in urban mobility and social outcomes.
- **UNDP Istanbul, SDG AI Lab** **Istanbul, Turkey (Remote)**
GIS Specialist (Part-time) *Dec 2020-Jul 2021*
 Developed LiDAR-based spatial impact-assessment tools for rapid flood-response modelling and natural-hazard risk assessment.

Education

- **University College London** **London, UK**
PhD in Advanced Spatial Analysis *2022-2027*
 In progress, thesis draft targeted around Oct 2026; official completion/conferral expected 2027. Dissertation: "Quantifying Data Injustice: Digital Geospatial Data for Urban Decision-Making in the Global South."
- **Imperial College London** **London, UK**
MSc in Environmental Technology, Merit *Sep 2019-Nov 2020*
 Thesis: "Distributed Ledger Technology-enabled passenger control for pandemic social distancing."
- **Shandong University** **Shandong, China**
BEng in Marine Resources Development Technology *Sep 2015-Jun 2019*

Languages

- Mandarin (Chinese) (native), English (full professional proficiency).

Selected Publications

- Zhang, W. et al. (2026). Identifying Digitally Invisible Groups: A Machine Learning Framework Using Multi-Source Data. *Transactions in Urban Data, Science, and Technology* ([DOI](#)).
- Zhang, W. et al. Visible Waste, Invisible Inequality: Mapping Sanitation Disparities from Street View Imagery in Nairobi. Manuscript under review, *Applied Geography* ([SSRN preprint](#)). Winner, Best Environmental and Climate Impact Paper, IDRC 2025.
- Zhang, W. et al. (2024). The NetMob2024 Dataset: Population Density and Origin-Destination Matrices from Four LMICs. [arXiv:2410.00453](#); World Bank Data Catalog dataset.
- Zhang, W. et al. (2025). Policy-Induced Push Migration: Measuring Cross-Border Mobility with Digital Trace Data. *MAPINEQ Research Report* ([report](#)); manuscript submitted to *International Migration Review*.
- Zhang, W., Alaa El-Din, K., Rapson, J., Nueno Guitart, J., Leasure, D.R. (2026). Gaza Tents: Automated Tent Mapping for Gaza using Satellite Imagery from 1 July 2026. *Real Good Research* ([report](#), [code](#)).

Funding and Awards

- **CSC Scholarship** (USD 21,600 + EUR 14,400).
- **UCL Global Engagement Fund** (£4,966) – co-secured and managed project funding and budget for a Nairobi stakeholder workshop with NGOs, APHRC and UN-Habitat.
- **Best Environmental and Climate Impact Paper** – 8th International Disaster and Risk Conference (IDRC 2025).
- **Research Accreditation**, Office for National Statistics (UK) – issued Jul 2023, expires Jul 2028.